CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the integers from -8 to 12 (both included) is: 1

(a) 42 **(b)** 52 **(c)** 41 **(d)** 43

2. On a number line, the distance between -7 and -23 is: 1

(a) 15 **(b)** 19 **(c)** 17 **(d)** 16

3. Two integers add to 8 and one of them is -9. The other is: 1

(a) 27 **(b)** 17 **(c)** 15 **(d)** 19

4. A trader loses ₹80 on day 1, gains ₹60 on day 2 and loses ₹90 on day 3. The net result in ₹ is: 1

(a) -100 **(b)** -110 **(c)** -112 **(d)** -108

5. The value of (-35) + 35 + (-9) is: 1

(a) -10 **(b)** 26 **(c)** -8 **(d)** -9

6. If ? + (-15) = -4, then the missing number ? is: 1

(a) 9 **(b)** 21 **(c)** 11 **(d)** 13

7. The value of -9 - 4 - 9 is: 1

(a) -12 **(b)** -20 **(c)** -24 **(d)** -22

8. The sum of the integers from -9 to 10 is: 1

(a) 19 **(b)** 9 **(c)** 10 **(d)** 11

9. **Assertion (A):** $-4 + 6 = 2$. 1

Reason (R): Subtracting a negative integer is the same as subtracting the corresponding positive integer.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The successor of -1 is -2. 1

Reason (R): The successor of an integer is obtained by adding 1 to it.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the missing number ?: $14 + ? = -6$. 2

12. A mountain climber starts at -45 m, goes up 100 m and then down 80 m. Find the climber's final height and its vertical distance from a camp at -10 m. 2

13. Find the sum of the integers from -12 to 15 inclusive. 2

14. Find the missing number ?: $-21 + ? = -8$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A submarine at -45 m dives 25 m, rises 60 m, then dives 15 m. Find its position after each move and the total vertical distance moved. 3

16. The sum of two integers is -15. One integer is -7. Find the other and their difference (larger minus smaller). 3

17. Write the next three terms of the pattern -30, -23, -16, -9, ... and find the first positive term. 3

OR

A submarine at -80 m dives 35 m, rises 20 m, then dives 45 m. Find its position after each move and the total vertical distance moved. 3

18. The sum of two integers is 20. One integer is 35. Find the other and their difference (larger minus smaller). 3

OR

Write the next three terms of the pattern -18, -13, -8, -3, ... and find the first positive term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A mall has floors below the ground (basements) numbered $-1, -2, \ldots$ and floors above the ground numbered $1, 2, \ldots$. A lift starts at the ground floor (0), goes down 3 floors, then up 5 floors, then down 4 floors. 5

(i) On which floor is the lift after the first move?

(ii) On which floor after the second move?

(iii) On which floor does it finish?

(iv) How many floors did the lift travel in all? Say which basement it finishes in.

20. At the start of a game a player has -3 points. In four rounds the points change by +8, -10, +4, -5. Find the score after each round, the final score, and the lowest score reached. 5

OR

A lift in a building with 3 basement floors starts at floor 2. It goes 4 floors down, 1 floors up, 4 floors down. State its final floor, and whether the final floor exists (the lowest basement is -3). 5

END OF QUESTION PAPER — Total: 40 Marks